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Sparse Bayesian Multidimensional Item Response Theory

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ABSTRACT

Multivariate Item Response Theory (MIRT) is sought-after widely by applied researchers looking for interpretable (sparse) explanations underlying response patterns in questionnaire data. There is, however, an unmet demand for such sparsity discovery tools in practice. Our article develops a Bayesian platform for binary and ordinal item MIRT which requires minimal tuning and scales well on large datasets due to its parallelizable features. Bayesian methodology for MIRT models has traditionally relied on MCMC simulation, which cannot only be slow in practice, but also often renders exact sparsity recovery impossible without additional thresholding. In this work, we develop a scalable Bayesian EM algorithm to estimate sparse factor loadings from mixed continuous, binary, and ordinal item responses. We address the seemingly insurmountable problem of unknown latent factor dimensionality with tools from Bayesian nonparametrics which enable estimating the number of factors. Rotations to sparsity through parameter expansion further enhance convergence and interpretability without identifiability constraints. In our simulation study, we show that our method reliably recovers both the factor dimensionality as well as the latent structure on high-dimensional synthetic data even for small samples. We demonstrate the practical usefulness of our approach on three datasets: an educational assessment dataset, a quality-of-life measurement dataset, and a bio-behavioral dataset. All demonstrations show that our tool yields interpretable estimates, facilitating interesting discoveries that might otherwise go unnoticed under a pure confirmatory factor analysis setting. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

Multidimensional Item Response Theory (MIRT) is indispensable to educational and psychological measurement research. It plays a dominant role in evaluating latent cognitive abilities of respondents, classifying subpopulations, and uncovering item characteristics (Bock and Gibbons 2021). MIRT models have been used to assess individual differences in personality traits and attitudes (Fraley, Waller, and Brennan 2000), improve health outcomes in clinical patient-reported outcome research (Forero et al. 2013), make scientific discoveries from intricate bio-behavioral data (Stan et al. 2020), and enhance testing efficiency and accuracy in computerized adaptive testing in behavioral health (Gibbons et al. 2016).

In the standard two-parameter MIRT framework (Bock and Gibbons 2021), we observe a binary matrix $Y \in \mathbb{R}^{N\times J}$, where each element Y_{ij} represents whether subject i answered item j correctly. Assuming the number of factors K is known, we denote the factor loading matrix as $B \in \mathbb{R}^{J\times K}$, the intercept as $D \in \mathbb{R}^{J}$, and subject i's latent trait as $\theta_i \in \mathbb{R}^{K}$. The two-parameter MIRT model aims to capture the following data generating process:

$$Y_{ij}|B_j, \theta_i, d_j \sim \text{Bernoulli}(g(B_j'\theta_i + d_j)),$$

for subjects $i = 1, \ldots, N$ and items $j = 1, \ldots, J$, where B_j, as a column vector, is the jth row of B, d_j is the jth element of the intercept D, and function $g(.)$ is the link function.

The most common approach to estimating MIRT models is based on the principle of marginal maximum likelihood (MML) estimation (Bock 1972). Following the development of the EM algorithm (Dempster, Laird, and Rubin 1977), Bock and Aitkin (1982) derive a computationally efficient procedure to estimate the item parameters using deterministic Gaussian quadrature and extend the EM principle to the exploratory MIRT setting. Another line of MIRT estimation roots in the idea of Monte Carlo Expectation Maximization algorithm (MCEM), in which the E-step is often replaced with Monte-Carlo methods (Meng and Schilling 1996). The popular Metropolis-Hastings Robbins-Monro (MHRM) algorithm performs stochastic imputation with the Metropolis-Hastings sampler in the E-step and Robbins-Monro stochastic approximation in the M-step (Cai 2010).

The Joint Maximum Likelihood (JML) estimation method has gained popularity for estimating latent variable models in large-scale discrete datasets. For example, a constrained JML estimator demonstrates remarkable scalability and effectiveness in estimating high-dimensional exploratory MIRT models (Chen, Li, and Zhang 2019). In a confirmatory context, Yunxiao Chen and Zhang (2020) proposed a variation of the JML estimation method within a generalized latent factor model framework, establishing the structural identifiability of latent factors under a double-asymptotic setting. Additionally, the JML estimation approach has been successfully extended to

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the discrete latent variable and the Cognitive Diagnostic Models (CDMs) framework (Gu and Xu 2023).

MCMC sampling coupled with data augmentation strategies of Albert and Chib (1993), or Pólya-Gamma variables (Polson, Scott, and Windle 2013; Jiang and Templin 2018) has long been the only resort for Bayesian statisticians to estimate MIRT models. However, Durante (2019) recently showed that for probit regression with a standard Gaussian prior on the coefficient parameter, the posterior follows a unified-skew normal distribution (Arellano-Valle and Azzalini 2006), which can be easily sampled from with the minimax tilting methods (Botev 2016). This discovery has made our derivation of the latent factor posterior distributions tractable under the probit factor model and motivates us to advocate for a Bayesian EM approach. Unlike the E-step in MCEM or MHRM algorithms which rely on MCMC sampling, our E-step sampling can be efficiently parallelized for each observation, and the M-step can also be decomposed into solving J independent penalized probit regressions, which can be solved by a standard `glmnet` estimation algorithm (Friedman, Hastie, and Tibshirani 2010).

By explicitly characterizing latent factor posterior distributions, this article develops a scalable Bayesian framework for estimating sparse exploratory MIRT models, with the aim of recovering an interpretable representation of the factor loading matrix B without making assumptions on its zero allocations. The process of finding a sparse solution can be facilitated by the idea of parameter expansion (Liu, Rubin, and Wu 1998), which enables rapid exploration of the parameter space of equivalent observed data likelihoods and is more robust against poor initialization. Our proposed framework and estimation procedures can also be further generalized beyond binary datasets to accommodate mixed responses. Parallel research in the CDM framework has also received significant attention recently: In CDMs, the latent structure, represented by the Q-matrix, is typically provided by experts. However, the Q-matrix can often be misspecified in practice. Consequently, sparsity recovery through the estimation of the Q-matrix has been explored in many recent works (Chen et al. 2015; de la Torre and Chiu 2016; Chen, Culpepper, and Liang 2020).

Besides its computational efficiency and versatility to accommodate mixed data type, one primary advantage of our Bayesian approach is its flexibility in prior specification. In many exploratory MIRT settings, the number of latent factors and the exact sparsity level of the loading matrix are typically unknown and are often misspecified. To incorporate our prior beliefs amid such uncertainty, we impose a hierarchical Indian Buffet Process (IBP) prior (Ghahramani and Griffiths 2005) coupled with the Spike-and-Slab LASSO (SSL) prior (Rockova 2018) on the factor loading matrix, following the idea in Ročková and George (2016). This allows the algorithm to automatically learn the latent dimensionality and produce sparse loading estimates. Additionally, we establish general identification conditions and provide theoretical guarantees for recovering the true number of latent factors and ensuring the consistency of the loading matrix. When the number of observations is relatively small, which is often the case in social science and behavior health research, our Bayesian approach can yield more reliable estimates by leveraging such prior information, resulting in a more sparse solution.

The rest of this article is organized as follows. Section 2 offers a concrete motivating example exhibiting the potential of our proposed approach to estimating sparse Bayesian MIRT models. Section 3 provides a detailed exposition of the data generation process, identification conditions, and the theoretical justifications behind our proposed hierarchical model. We then show the derivations of our parameter-expanded EM algorithm in Section 4. Section 5 further demonstrates the generality of our proposed methodology by accommodating mixed data types. In Section 6, the effectiveness of our proposed method is substantiated through its application to diverse scenarios: a challenging synthetic dataset, a large sample educational assessment dataset, a quality-of-life measurement dataset, and a challenging ordinal bio-behavioral dataset. We conclude the article in Section 7 and provide additional details in the Appendix.

2. Exact Sparsity Recovery

Prior to delving into technical details, we provide a simple motivating example to illustrate the usefulness of our Bayesian approach when the factor loading matrix is sparse and its dimensionality uncertain. To this end, we generated four-factor, two-parameter binary item response data, and focused on a challenging high-dimensional $N < J$ case, in which 50% of items load on an arbitrary pair of latent factors. As shown in the leftmost subplot in Figure 1, the true loading matrix is relatively sparse—the light pink areas represent zeros, while the red areas represent ones. Although simple looking, this loading structure can be very hard to recover in practice, as there exists a substantial overlap among latent factors.

For benchmarking, we consider the other three state-of-the art exploratory MIRT methods, where no assumptions are made for the zero allocations of the loading matrix : (a) the MHRM algorithm (Cai 2010), (b) a customized Gibbs sampler with an adaptive spike-and-slab normal prior (Ishwaran and Rao 2005) (derivations in Appendix A), and (c) the exploratory JML estimation method (Chen, Li, and Zhang 2019). The MHRM and the JML method implementations are available in the `mirt` and the `mirtjml` R packages (Chalmers 2012).

To demonstrate that our approach is capable of learning the dimensionality and of producing a sparse solution, we purposefully mis-specified (overshot) the number of factors $K^* = 10$ for our EM algorithm. As shown in Figure 1, our PXL-EM algorithm recovered the true overlapping latent structure, and correctly identified the true number of factors as 4, despite us purposefully overshooting the number of factors. For the MHRM algorithm, even given the true number of factors as 4, its estimated loading matrix remained structureless.

To fit our customized sparse Gibbs sampler, we lower triangularized the factor loading matrix for identification purpose (Aguilar and West 2000). While the Gibbs sampler identified four prominent latent factors with stripe-like structure, it did not recover the substantial overlapping components among the factors, and was incapable of outputing exact zero values. For the JML estimation method, although it converges significantly faster compared to the other approaches, it only identified three factors and failed to recover the overlapping components.

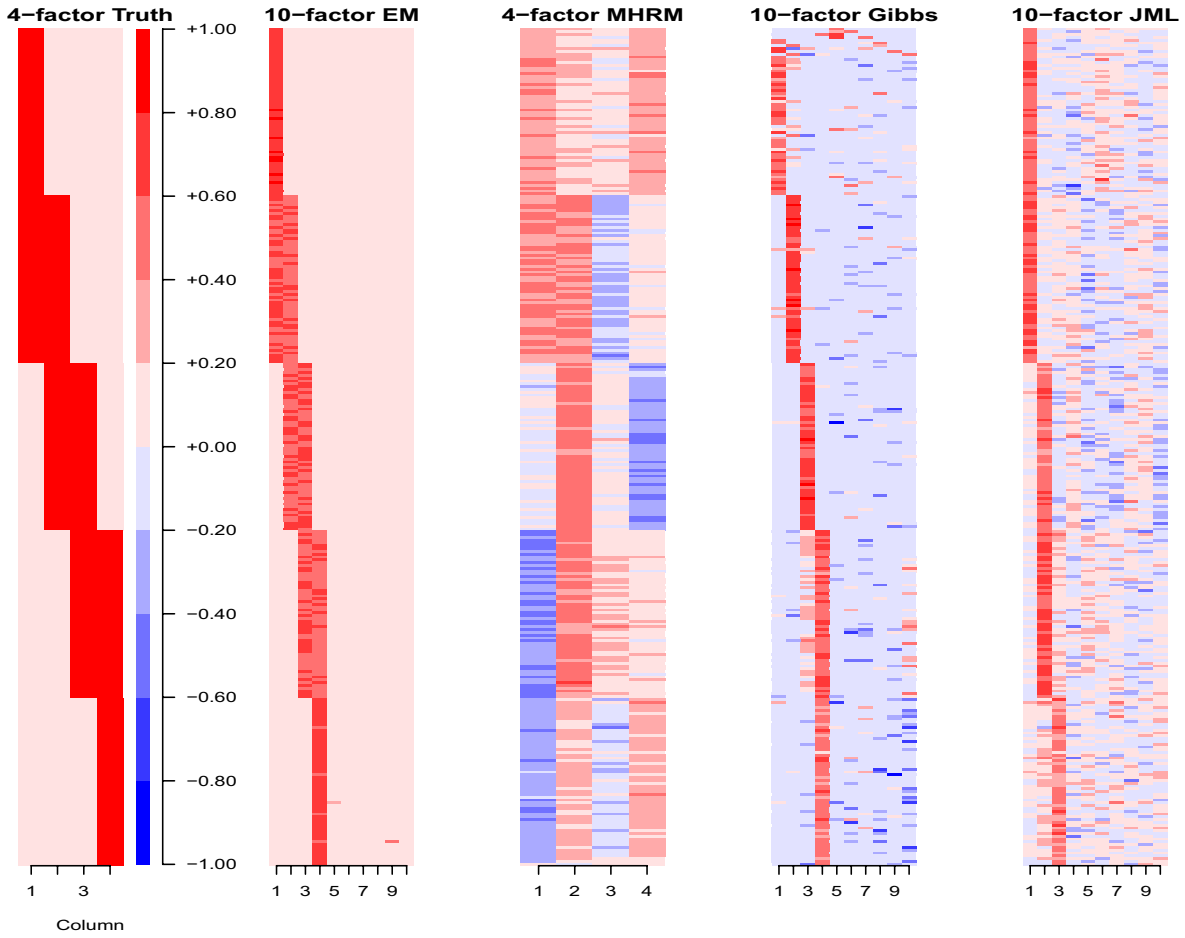


Figure 1. Factor loading matrix estimation comparison.

This simple motivating example effectively highlights the reasoning behind our endorsement of the Bayesian EM approach to MIRT over alternatives.

3. Nonparametric Bayesian MIRT

We draw inspiration from Durante's recent work (Durante 2019), which characterizes the posterior distribution of coefficient parameters in a standard probit regression. This insight guides us toward a standard normal cumulative distribution link function $\Phi(.)$ to model item response data. For simplicity, we start by considering the binary case:

$$Y_{ij}|\theta_i, B_j, d_j \sim \text{Bernoulli}(\Phi(B_j'\theta_i + d_j)). \tag{1}$$

Building upon the Bayesian factor analysis framework introduced by Ročková and George (2016), we deploy the spike-and-slab LASSO (SSL) prior (Rockova 2018) on the loading matrix to prompt sparsity. The sparsity assumption is essential, as it alleviates model identification issues by anchoring estimation on sparse solutions, leading to more interpretable outcomes. Additionally, the SSL prior reflects many real-world applications where factor loadings are inherently sparse, and therefore has the potential to improve out-of-sample prediction. Our SSL approach employs a two-point mixture of Laplace distributions which can adaptively produce exact zeros, and does not necessitate an additional thresholding step. For each element B_{jk} in the loading matrix, we couple the SSL prior with a binary variable selection variable $\gamma_{jk} \in \{0, 1\}$. Let $\lambda_{0k} >> \lambda_1 > 0$, and let $\psi(\beta|\lambda) = \frac{\lambda}{2}\exp\{-\lambda|\beta|\}$ denote the Laplace prior with mean 0 and variance $\frac{2}{\lambda^2}$. We consider the following hierarchical prior:

$$\pi(B_{jk}|\gamma_{jk}, \lambda_{0k}, \lambda_1) \sim (1-\gamma_{jk})\psi(B_{jk}|\lambda_{0k}) + \gamma_{jk}\psi(B_{jk}|\lambda_1).$$

To avoid committing to a fixed number of factors, we deploy the infinite-dimensional Indian Buffet Process (IBP) prior with intensity parameter $\alpha > 0$ on the sparsity pattern of the loading matrix B. A larger α value would favor a less sparse loading matrix estimation. This prior serves as variable selection tool and accounts for uncertainty in the latent space dimensionality. As observed in Ročková and George (2016), posterior simulation can be greatly facilitated if we consider the stick-breaking representation of IBP (Teh, Görür, and Ghahramani 2007). Consequently, researchers simply need to choose a large enough truncation-level K^* as an upper bound for the number of latent factors. In the case where estimated loading matrix contains no sparse columns, one should consider increasing K^* to avoid underestimation of the true number of factors. For each item $j = 1, \ldots, J$, and each latent dimension $k = 1, \ldots, K^*$, the stick-breaking representation of our IBP prior can be expressed as follows:

$$v_l \overset{\text{iid}}{\sim} \text{Beta}(\alpha, 1),\ c_k = \prod_{l=1}^{k} v_l,\ \gamma_{jk}|c_k \sim \text{Bernoulli}(c_k).$$

The data generation process is finalized upon specifying priors for both the latent factor θ_i and the intercept d_j. In the context of exploratory factor analysis, it is customary to assume a prior of zero-mean multivariate Gaussian with an identity covariance matrix for the latent factors, since our goal is to unveil the latent structure of factor loadings. We further consider a Laplacian prior with small parameter λ_1 for the intercept parameter d_j. The combination of SSL and IBP priors on the loading matrix enhances the flexibility and interpretability of our model, which also has exhibited notable success in the nonparametric linear factor model settings (Ročková and George 2016).

3.1. Probit Identifiability

One of the key advantages of adopting a probit link $\Phi(.)$ is its ability to simplify the establishment of identifiability results in categorical factor analysis. In contrast, the convolution of Gaussian and logistic random variables induced by a logistic link often renders many classic proof techniques within linear factor analysis intractable. As observed in Fang et al. (2021), a necessary and sufficient condition to check the identifiability of a probit factor model is to examine the tetrachoric correlations as follows:

Proposition 3.1. (Fang et al. 2021) Let $A, B \in \mathbb{R}^{J\times K}$ be the factor loading matrices, $D_1, D_2 \in \mathbb{R}^J$ be the intercepts, and the latent factor θ be the zero-mean multivariate Gaussian with covariance matrix Σ. Then two sets of parameters (A, D_1) and (B, D_2) define the same probit factor model if and only if their thresholds and tetrachoric correlations are equal:

$$\frac{D_{1j}}{\sqrt{A_j'\Sigma A_j + 1}} = \frac{D_{2j}}{\sqrt{B_j'\Sigma B_j + 1}} \quad \forall j, \tag{2}$$

$$\frac{A_{j_1}'\Sigma A_{j_2}}{\sqrt{A_{j_1}'\Sigma A_{j_1} + 1}\sqrt{A_{j_2}'\Sigma A_{j_2} + 1}} = \frac{B_{j_1}'\Sigma B_{j_2}}{\sqrt{B_{j_1}'\Sigma B_{j_1} + 1}\sqrt{B_{j_2}'\Sigma B_{j_2} + 1}} \quad \forall j_1 \neq j_2. \tag{3}$$

By Proposition 3.1, establishing identifiability of a given probit factor model is equivalent to showing that equations 2 and 3 induce a unique set of model parameters (B, D). Recall in a linear setting, one needs to show $AA' + \Sigma_1 = BB' + \Sigma_2$ yields the unique set of model parameters (B, Σ), where Σ is a diagonal matrix of variances. Though linear identifiability results are not directly applicable under the probit link, the presence of a comparable equation structure suggests the possibility of adapting classical linear identifiability principles to the probit factor model setting. In particular, Fang et al. (2021) leverages Proposition 3.1 to establish identifiability conditions for confirmatory probit bifactor and two-tier models.

In the exploratory linear factor analysis setting, the most common strategy to resolve the rotational invariance identifiability issue is to constrain the loading matrix to possess of a positive lower triangularized (PLT) form (Geweke and Zhou 1996; West 2003). More recently, Frühwirth-Schnatter, Hosszejni, and Lopes (2023) establishes linear factor model identifiability results with generalized lower triangular (GLT) structure.

Definition 3.2. For factor loading matrix $B \in \mathbb{R}^{J\times K}$ with full column rank K and columns $k \in \{1, \dots K\}$, we define the *pivot row* l_k as the row index of the first nonzero factor loading in column k, such that $B_{i,k} = 0$ for all $i < l_k$. We say B has an *ordered GLT* structure if $l_1 < \cdots < l_K$, and $B_{l_k,k} > 0$ for all k.

First observe that the popular positive lower triangular loading constraint is an instance of the GLT structure. As suggested in Anderson and Rubin (1956) and Frühwirth-Schnatter, Hosszejni, and Lopes (2023), while PLT and GLT structures are effective in resolving rotational indeterminacy, they alone are insufficient for identifying linear factor models unless the variance matrix Σ is also precisely determined. For our proposed probit MIRT model, we first provide a sufficient identification condition for the intercept D, analogous to Theorem 5.1 of Anderson and Rubin (1956), and then establish similar sufficient identifiability condition for the loading matrix B under the GLT structures, provided that the intercept D is uniquely identifiable. The proof is available in Appendix B.1.

Theorem 3.3. Consider a probit factor model with a full column rank factor loading matrix $B = [B_1, \dots, B_J]' \in \mathbb{R}^{J\times K}$ and intercept $D \in \mathbb{R}^J$. Define matrix $\tilde{B} := [\tilde{B}_1, \dots, \tilde{B}_J]'$, where $\tilde{B}_j = \frac{B_j}{\sqrt{\|B_j\|^2+1}}$. Then:

A. A sufficient condition for identification of D is that if any row of $\tilde{B}$ is deleted, there remain two disjoint submatrices of rank K.
B. An ordered GLT structure is uniquely identified, provided that the intercept D is identified.

As noted in Frühwirth-Schnatter, Hosszejni, and Lopes (2023), the identifiability conditions for linear factor models usually involve two stages (Anderson and Rubin 1956): the identification of factor loading matrix (hindered by rotational indeterminacy) and the identification of variance, though the majority of statistics literature tends to focus on the former case. Theorem 3.3 hints the proofs of probit model identifiability results are likely to follow a similar process as in a linear factor setting, where variance identification is replaced by the intercept identification.

Once we know the factor loading matrix with the GLT structure is uniquely identified, a natural question is whether we can achieve posterior consistency of B under such a structural assumption. Notably, recent work by Ma and Liu (2020) advocates for a "$\sqrt{n}$-orthonormal factor model" approach under the SSL-IBP prior linear factor model setup, and establishes consistency of B up to sign permutation, condition on the feature allocation matrix Γ possessing the GLT structure. Their approach can also help address the "magnitude inflation problem" in high-dimensional setting, effectively mitigating sensitivity to the prior.

3.2. Posterior Asymptotics

We provide robust theoretical justification for our proposed nonparametric MIRT model, demonstrating its capability to recover the true number of factors and achieving the desired posterior contraction rate for the factor loading matrix. The

effective dimension of the loading matrix $K(B)$ under the SSL-IBP($\{\lambda_{0k}, \lambda_1, C\}$) prior is defined as the largest column index K such that $|B_{jk}| < \frac{1}{\lambda_{0k}-\lambda_1}\log[\frac{\lambda_{0k}(1-c_k)}{\lambda_1 c_k}]$ for any $j \in \{1,\dots,J\}$ and $k > K$ (Ročková and George 2016). We denote $B_{0n} \in \mathbb{R}^{J_n \times \infty}$ as the true loading matrix with k_{0n} nonzero columns, $\Theta_{0n} \in \mathbb{R}^{n\times\infty}$ as the latent factor matrix, and $\eta_{0ij} = B'_{0j}\theta_{0i}$. For $1 \le k \le k_{0n}$, let $S_{nk} = \sum_{j=1}^{J_n} \mathbb{I}(B_{jk} \neq 0)$, $\bar{S}_n = \frac{1}{k_{0n}}\sum_{k=1}^{k_{0n}} S_{nk}$, and assume each column of B_{0n} exhibits S_n-sparsity.

To study the asymptotic properties of the posterior, we make the following assumptions: (A) $n < J_n$ and $\lim_{n\to\infty}\sqrt{\frac{S_n k_{0n}\log^2 J_n}{n}} = 0$; (B) $\|B_{0n}\|_2 < \sqrt{\bar{S}_n}$, where $\|B_{0n}\|_2$ is the spectral norm of the submatrix formed by the first k_{0n} columns $\|B_{0n,k_{0n}}\|_2$; and (C) $\gamma_n(B_{0n},\Theta_{0n}) \precsim \log J_n$, where ϕ is standard normal PDF, and

$$\gamma_n(B_{0n},\Theta_{0n}) := 1 + \max_{\{i\le n, j\le J_n\}} \frac{\phi^2(\eta_{0ij})}{\Phi(\eta_{0ij})(1-\Phi(\eta_{0ij}))}.$$

Assumptions A and B are similar to the assumptions made in Ročková and George (2016), in which A restricts our analysis on the high-dimensional case, and B holds with high probability (Ročková and George 2016). Assumption C is an additional sparsity assumption on (B_{0n},Θ_{0n}), analogous to a special case of assumption 1 appeared in Jeong and Ghosal (2020) under the sparse generalized linear model setting. Given that $\frac{\phi^2(\eta_{0ij})}{\Phi(\eta_{0ij})(1-\Phi(\eta_{0ij}))}$ grows linearly in η_{0ij} by the Mills ratio while $\max_{i,j}\eta_{0ij}$ roughly grows in the log scale, Assumption C is relatively mild.

Theorem 3.4 shows the average posterior probability of our exploratory MIRT model overestimating the true number of factors, scaled by the true sparsity level, approaches zero asymptotically. This result is analogous to Theorem 3 in Ročková and George (2016) within a linear setting.

Theorem 3.4. Under assumptions (A)-(C), consider the sparse Bayesian MIRT model defined in (1), where the true loading matrix B_{0n} has k_{0n} nonzero leftmost columns. For $1 \le k \le k_{0n}$, let $\sum_{j=1}^{J_n}\mathbb{I}(B_{jk} \neq 0) = S_{nk}$ and $\lambda_{0k} \ge \frac{2J_n^2 k_{0n}^3 n}{S_{nk}}$. Then with the prior choice $B \sim$ SSL-IBP($\{\lambda_{0k}\}, \lambda_1, \alpha_n$), where $\alpha = \frac{1}{J_n}$, $\lambda_1 < e^{-2}$, and with some constant $C_0 > 0$:

$$E_{B_0} P[K(B) > C_0 k_{0n} S_n | Y^{(n)}] \xrightarrow[n\to\infty]{} 0.$$

The corresponding proof can be found in Appendix B.2. In the linear factor model setting (Ročková and George 2016), one may easily integrate Gaussian factor over the likelihood to study the posterior distribution of the covariance matrix. This approach is no longer possible under the probit link. Consequently, we introduce assumption (C) to control the Kullback-Leibler divergence and its variation between the posterior density and the true density.

The following theorem demonstrates the posterior distribution of the factor loading matrix B contracts toward the true factor loading matrix B_0 in terms of the root-average-squared Hellinger distance. For any two loading matrices B_1 and B_2, recall the average of the squares of the Hellinger distances is defined as $H_n^2(B_1,B_2) = \frac{1}{n}\sum_{i=1}^n H^2(f_{B_1,i}, f_{B_2,i})$, where $H^2(f_{B_1,i}, f_{B_2,i}) = \int(\sqrt{f_{B_1,i}} - \sqrt{f_{B_2,i}})^2 d\nu$ represents the squared Hellinger distance between the densities of the binary vector Y_i induced by B_1 and B_2, respectively.

Theorem 3.5. Assuming the true loading matrix B_{0n} has k_{0n} columns, each of which is S_n-sparse. Under assumptions (A)–(C), and with the same SSL-IBP prior choice as in Theorem 3.4, these exists a constant C_0' such that

$$E_{B_0}\Pi_n\left\{B : H_n(B,B_0) > C_0'\sqrt{\frac{k_{0n}S_n\log^2 J_n}{n}} | Y^{(n)}\right\} \xrightarrow[n\to\infty]{} 0.$$

The Hellinger posterior contraction rate for the regression parameter in generalized linear models with a fixed design was recently shown to be $\sqrt{\frac{S_0\log J_n}{n}}$, where S_0 is the size of the true support (Jeong and Ghosal 2020). Theorem 3.5 provides similar contraction guarantee but with an extra $\sqrt{\log J_n}$ factor, reflecting the cost of the growing number of factors k_{0n} and the support size S_n. The proof of Theorem 3.5 is detailed in Appendix B.2. In Appendix E.2, we further empirically verify the consistency of B with a challenging high-dimensional example as described in Section 6.2.

4. Parameter-Expanded EM for Sparse MIRT

Our proposed EM algorithm is guided by a recent groundbreaking result from (Durante 2019), who fully characterized the posterior distribution of coefficients in the standard probit regression model. This result enabled us to explicitly derive the latent factor posterior distributions, resulting in highly efficient E-step sampling.

4.1. Algorithm Setup

For simplicity, we let K represent the truncated level of the IBP prior K^*. Additionally, let $\Theta = [\theta_1,\dots,\theta_N]' \in \mathbb{R}^{N\times K}$ be the matrix of latent factors, $C = [c_1,\dots,c_k] \in \mathbb{R}^K$ be the vector of ordered inclusion probabilities, $D = [d_1,\dots,d_J]' \in \mathbb{R}^J$ be the intercept parameter, $B \in \mathbb{R}^{J\times K}$ be the factor loading matrix, and $\Gamma \in \mathbb{R}^{J\times K}$ be the binary latent allocation matrix. The goal of our EM algorithm is to estimate the model parameters $\Delta := (B,C,D)$. Given the initialization $\Delta^{(0)}$, the $(m+1)$st step of the EM algorithm finds $\Delta^{(m+1)}$ by solving the following optimization problem:

$$\begin{aligned}\Delta^{(m+1)} &= \text{argmax}_\Delta \mathbb{E}_{\Gamma,\Theta|Y,\Delta^{(m)}}[\log\pi(\Delta,\Gamma,\Theta|Y))] \\ &:= \text{argmax}_\Delta Q(\Delta).\end{aligned}$$

Let $\langle X\rangle$ represent $\mathbb{E}_{\Gamma,\Theta|Y,\Delta}(X)$, the conditional expectation given the observed data Y and model parameters Δ. Due to the hierarchical structure of our model, the optimization problem can be simplified by recognizing that the model parameters (B,D) and C are conditionally independent given (Γ,Θ). Thus, we can decompose $Q(\Delta)$ into the sum of $Q_1(B,D)$ and $Q_2(C)$, where

$$\begin{aligned}Q_1(B,D) &\propto \langle\log\pi(Y|B,D,\Theta)\rangle + \langle\log\pi(B|\Gamma)\rangle + \langle\log\pi(D)\rangle, \\ Q_2(C) &\propto \log(\langle\Gamma\rangle|C) + \log\pi(C).\end{aligned}$$

In the MIRT literature, it is customary to assume that item responses for each individual are conditionally independent given their multidimensional latent traits. This assumption enables us to write $Q_1(B, D)$ as $\sum_{j=1}^{J} Q_j(B_j, d_j)$, where

$$Q_j(B_j, d_j) = \sum_{i=1}^{N} \langle \log \Phi((2Y_{ij} - 1)(B_j'\theta_i + d_j)) \rangle - \sum_{k=1}^{K} |B_{jk}|(\lambda_1 \langle \gamma_{jk} \rangle + \lambda_0(1 - \langle \gamma_{jk} \rangle)) - \lambda_1 |d_j|. \tag{4}$$

This highlights the primary components of our EM algorithm. In the E-step, we compute the conditional expectations $\langle \log \Phi((2Y_{ij} - 1)(B_j'\theta_i + d_j)) \rangle$ and $\langle \gamma_{jk} \rangle$. In the M-step, we separately maximize $Q_1(B, D)$ and $Q_2(C)$. Efficient maximization of $Q_1(B, D)$ is achievable by decomposing it into J distinct objective functions, allowing for parallel optimization.

4.2. The E-Step

The complexity of the above-presented EM algorithm primarily stems from the computation of the conditional expectation term $\langle \log \Phi((2Y_{ij} - 1)(B_j'\theta_i + d_j)) \rangle$, given the challenge of identifying latent factor posterior distributions. In Bayesian linear factor model (Ročková and George 2016), one can interchange the expectation and the linear functional form, thereby enabling the computation of the posterior distribution $\langle \theta_i \rangle$ through a form of ridge regression. However, in our MIRT setting, relocating the expectation $\langle . \rangle$ within the log $\Phi(.)$ function would lead to maximizing an upper bound, constrained by the Jensen's inequality, and would not be valid even in an approximation sense.

4.2.1. Characterize Posterior Distribution of Latent Factors

To compute $\langle \log \Phi((2Y_{ij} - 1)(B_j'\theta_i + d_j)) \rangle$, we propose to directly sample from the posterior distribution of θ_i and perform Monte Carlo integration. This strategy can yield significantly enhanced computational speed when compared to MCEM-type algorithms relying on MCMC sampling. The advantage of our proposed E-step becomes particularly prominent when dealing with a large number of Monte Carlo samples in high latent dimensions, as MCMC sampling is not amenable to parallelization, thereby resulting in increased computational costs. The efficient sampling of latent factors is facilitated by a recent result by Durante (2019), who derives the posterior distribution of coefficient parameters in a probit regression as the unified skew-normal distribution (Arellano-Valle and Azzalini 2006), which can be sampled efficiently:

Definition 4.1. Let $\Phi_J\{V; \Sigma\}$ represent the cumulative distribution functions of a J-dimensional multivariate Gaussian distribution $N_J(0_J, \Sigma)$ evaluated at vector V. A K-dimensional random vector $z \sim \text{SUN}_{K,J}(\xi, \Omega, \Delta, \gamma, \Gamma)$ has the unified skew-normal distribution if it has the probability density function:

$$\phi_K(z - \xi; \Omega) \frac{\Phi_J\left\{\gamma + \Delta^{\mathrm{T}} \bar{\Omega}^{-1} \omega^{-1}(z - \xi); \Gamma - \Delta^{\mathrm{T}} \bar{\Omega}^{-1} \Delta\right\}}{\Phi_J(\gamma; \Gamma)}.$$

Here, $\phi_K(z - \xi; \Omega)$ is the density of a K-dimensional multivariate Gaussian with expectation $\xi = (\xi_1, \ldots, \xi_K)'$, and a K by K covariance matrix $\Omega = \omega \bar{\Omega} \omega$, where $\bar{\Omega}$ is the correlation matrix and ω is a diagonal matrix with the square roots of the diagonal elements of Ω in its diagonal. Δ is a K by J matrix that determines the skewness of the distribution, and $\gamma \in \mathbb{R}^J$ control the flexibility in departures from normality.

In addition, the $(K + J) \times (K + J)$ matrix Ω^*, having blocks $\Omega^*_{[11]} = \Gamma, \Omega^*_{[22]} = \bar{\Omega}$ and $\Omega^*_{[21]} = \Omega^{*'}_{[12]} = \Delta$, needs to be a full-rank correlation matrix.

In the context of our EM algorithm, a key difference from Durante (2019) rises in that the intercept D is held fixed during the E-step, and each d_j is distinct for every item. Fortunately, it remains feasible to deduce the posterior distribution of θ_i as an instance of the unified skew normal distribution:

Theorem 4.2. Suppose the factor loading matrix B and the intercept D are known. If $y = (y_1, \ldots, y_J)' \in \mathbb{R}^J$ is conditionally independent binary response data from the probit MIRT model defined in (1), and $\theta_i \sim N(\xi, \Omega)$ assumed to be K-dimensional Gaussian prior, then

$$(\theta_i \mid y, B, D) \sim \text{SUN}_{K,J}\left(\xi_{\text{post}}, \Omega_{\text{post}}, \Delta_{\text{post}}, \gamma_{\text{post}}, \Gamma_{\text{post}}\right),$$

with posterior parameters

$$\begin{aligned} \xi_{\text{post}} &= \xi, \quad \Omega_{\text{post}} = \Omega, \quad \Delta_{\text{post}} = \bar{\Omega} \omega D_1^{\mathrm{T}} S^{-1}, \\ \gamma_{\text{post}} &= S^{-1}(D_1 \xi + D_2), \\ \Gamma_{\text{post}} &= S^{-1}\left(D_1 \Omega D_1^{\mathrm{T}} + \mathbb{I}_J\right) S^{-1}, \end{aligned}$$

where $D_1 = \text{diag}(2y_1 - 1, \ldots, 2y_J - 1)B$ with jth row as D_{1j}, $D_2 = \text{diag}(2y_1 - 1, \ldots, 2y_J - 1)D \in \mathbb{R}^J$, and $S = \text{diag}\{(D_{11}' \Omega D_{11} + 1)^{\frac{1}{2}}, \ldots, (D_{1J}' \Omega D_{1J} + 1)^{\frac{1}{2}}\} \in \mathbb{R}^{J \times J}$.

Theorem 4.2 is useful as we can now entirely characterize the latent factor posterior distribution. The significance of it goes beyond leading to a tractable E-step in our proposed algorithm, as it has direct applications to psychological measurements: given any J well-calibrated items and for any new observation $y \in \mathbb{R}^J$, Theorem 4.2 immediately provides an analytic form of the latent trait posterior distribution. In consequence, no extra Newton step or MCMC simulation are needed for latent factor inference.

Given the latent factor posterior distributions, the E-step computation becomes straightforward via Monte Carlo integration. As the posterior distribution of θ_i is not identical to that of a probit regression, we can instead consider the following sampling strategy:

Corollary 4.3. If $\theta_i | y, B, D$ has the unified skew-normal distribution from Theorem 4.2, then

$$(\theta_i | y, B, D) \overset{\text{d}}{=} \xi + \omega \left\{ V_0 + \bar{\Omega} \omega D_1' \left(D_1 \Omega D_1' + \mathbb{I}_{\mathbb{J}}\right)^{-1} S V_1 \right\}, \quad (V_0 \perp V_1),$$

with $V_0 \sim N\left\{0, \bar{\Omega} - \bar{\Omega} \omega D_1' \left(D_1 \Omega D_1' + \mathbb{I}_{\mathbb{J}}\right)^{-1} D_1 \omega \bar{\Omega}\right\}$, and V_1 from a zero mean J-variate truncated normal with covariance matrix $S^{-1}\left(D_1 \Omega D_1' + \mathbb{I}_{\mathbb{J}}\right) S^{-1}$ and truncation below $-S^{-1}(D_1 \xi + D_2)$.

The proofs of Theorem 4.2 and Corollary 4.3 can be found in Appendix C.

4.2.2. E-step Computation

Corollary 4.3 provides us with a direct approach to assess $\langle \log \Phi((2Y_{ij}-1)(B_j'\theta_i + d_j))\rangle$. This involves sampling a sufficiently large number of θ_i values from its posterior distribution and subsequently performing Monte Carlo integration. Assuming we require M samples for each latent factor θ_i, we can generate individual samples $\theta_i^{(m)}, m = 1,\dots,M$, by following these three steps:

- Step 1: Sample $V_0^{(m)} \sim N\left\{0, \bar{\Omega} - \bar{\Omega}\omega D_1'\left(D_1\Omega D_1' + \mathbb{I}_{\mathbb{J}}\right)^{-1} D_1\omega\bar{\Omega}\right\} \in \mathbb{R}^K$.
- Step 2: Leveraging the minimax tilting methods of Botev (2016), sample $V_1^{(m)}$ from a zero mean J-variate truncated normal with covariance matrix $S^{-1}\left(D_1\Omega D_1' + \mathbb{I}_{\mathbb{J}}\right)S^{-1}$ and truncation below $-S^{-1}(D_1\xi + D_2)$.
- Step 3: Compute $\theta_i^{(m)} = \xi + \omega\left\{V_0^{(m)} + \bar{\Omega}\omega D_1'\left(D_1\Omega D_1' + \mathbb{I}_{\mathbb{J}}\right)^{-1} SV_1^{(m)}\right\}$.

Finally, we need to compute the conditional distribution $\langle\gamma_{jk}\rangle$ as

$$\langle\gamma_{jk}\rangle = P(\gamma_{jk}=1|\Delta) = \frac{c_k\psi(B_{jk}|\lambda_1)}{c_k\psi(B_{jk}|\lambda_1) + (1-c_k)\psi(B_{jk}|\lambda_0)}.$$

4.3. The M-step

As discussed in Section 4.1, the M-Step of our algorithm can be decomposed into two independent optimization problems. To maximize $Q_1(B,D)$, we can maximize each $Q_j(B_j,d_j)$ as defined in (4) in parallel. Suppose we obtain M samples of $(\theta_i^{(1)},\dots,\theta_i^{(M)})$ from the E-Step for each subject i, we can maximize the following surrogate objective function:

$$\frac{1}{M}\sum_{i=1}^{N}\sum_{m=1}^{M}\log\Phi((2Y_{ij}-1)(B_j'\theta_i^{(m)} + d_j)) - \sum_{k=1}^{K}|B_{jk}|(\lambda_1\langle\gamma_{jk}\rangle + \lambda_0(1-\langle\gamma_{jk}\rangle)) - \lambda_1|d_j|.$$

Hence, maximizing $Q_j(B_j,d_j)$ is equivalent to estimating a penalized probit regression with an intercept. Specifically, this regression involves $N \times M$ response variables, which result from replicating the jth column of response Y_j for M times. The corresponding design matrix consists of M identical copies of the latent factor matrix Θ. For each latent dimension k, we apply a distinct l_1 penalty to B_{jk}, with a weight of $(\lambda_1\langle\gamma_{jk}\rangle + \lambda_0(1-\langle\gamma_{jk}\rangle))$. Solving this penalized regression can be efficiently achieved using standard off-the-shelf packages, such as the "glmnet" package (Friedman, Hastie, and Tibshirani 2010) in R.

To maximize $Q_2(C)$, we need to solve the following constraint linear program:

$$\max_{C} \sum_{j=1}^{J}\sum_{k=1}^{K}[\langle\gamma_{jk}\rangle \log c_k + (1-\langle\gamma_{jk}\rangle)\log(1-c_k)] + (\alpha-1)\log(c_K)$$
$$\text{s.t.} \quad c_k - c_{k-1} \le 0, k = 2,\dots,K,$$
$$0 \le c_k \le 1, k = 1,\dots,K.$$

Note the optimization problem is convex, and hence can be solved with standard convex optimization package. As observed in Ročková and George (2016), the optimization process can be accelerated if we sort the factors by their binary order before optimizing the problem.

4.4. The PXL-EM Algorithm

We present the parameter-expanded version of our EM algorithm (PXL-EM) to expedite convergence and enhance exploration of the parameter space. The updates for our PXL-EM algorithm are almost as identical to vanilla EM algorithm described above, except we now need an additional auxiliary parameter A, which serves to rotate the loading matrix B toward a sparser configuration (PXL-EM). A brief summary of our PXL-EM algorithm can be found in the table of Algorithm 1. The derivation and the justification of the factor rotation step is provided in Appendix D.

5. Extension to Mixed Data Types

We demonstrate the extension of our proposed methodology, illustrated for binary datasets, to item response with mixed data type. For each observation $Y_i \in \mathbb{R}^J$, we allow Y_{ij} to be categorized as continuous, binary, or ordinal. Specifically, let J_C be the set of continuous responses, J_B be the set of binary responses, J_O be the set of ordinal responses, then we have $J = J_B \cup J_C \cup J_O$. let $B_S \in \mathbb{R}^{|S|\times K}$ be the sub matrix of B indexed by set S. Assuming the binary responses $j \in J_B$ are generated by the standard MIRT model as defined in (1), we consider the following data generating process for the continuous and ordinal items:

- For each continuous feature $j \in J_c$, we assume it has been centered, and assign a noninformative inverse gamma prior on its unknown variance $\sigma_j^2 \sim \text{IG}(\eta_1,\eta_2)$. Then
$$Y_i|\theta_i, B_{J_c}, \Sigma \sim N(B_{J_c}\theta_i, \Sigma), \quad \Sigma := \text{diag}(\{\sigma_j^2\}_{j\in J_c}). \tag{5}$$
- For each ordinal feature j that takes values in $\{0,1,\dots,L_j\}$, we assume it has been generated from the standard graded response model Samejima (1968):
$$P(Y_{ij}=l|\theta_i,B_j,d_j) = P(Y_{ij}\le l) - P(Y_{ij}\le l-1) = \Phi(B_j'\theta_i + d_{j,l}) - \Phi(B_j'\theta_i + d_{j,l-1}), \tag{6}$$
where we need to estimate L_j-dimensional intercept d_j such that $-\infty = d_{j,-1} < d_{j,0} \le \cdots \le d_{j,L_j-1} < d_{j,L_j} = \infty$.

To derive a tractable E-step, we will show the latent factor posterior distribution still belongs to an instance of SUN under the mixed item response types. For ease of notation, we fix an observation i and write $Y := Y_i \in \mathbb{R}^J$. Define J_{O_1} as the set of ordinal features for which $y_{ij} \notin \{0, L_j\}$, and $J_{O_2} = J_O \setminus J_{O_1}$. Let D_{J_B} and Y_{J_B} denote the sub vectors of D and Y indexed by J_B, respectively. We further define the following quantities:

Algorithm 1: PXL-EM Algorithm for Sparse MIRT

Data: Item Response $Y \in \mathbb{R}^{N\times J}$
Initialize model parameters $B = B^{(0)}, C = C^{(0)}, D = D^{(0)}$;
while *PXL-EM has not converged* **do**
  /* E-Step */
  for $i = 1$ **to** N **do**
    Draw $\theta_i^{(1)}, \ldots, \theta_i^{(M)}$ as described in Section 4.2.2 ; /* Update Latent Factors */
  end
  $\langle \gamma_{jk} \rangle = \frac{c_k \psi(B_{jk}|\lambda_1)}{c_k \psi(B_{jk}|\lambda_1)+(1-c_k)\psi(B_{jk}|\lambda_0)}$; /* Update Latent Indicators */
  /* M-Step */
  for $j = 1$ **to** J **do**
    $B_j, d_j = \arg\max Q_j(B_j, d_j)$, as described in Section 4.3 ; /* Update Loadings */
  end
  $C = \arg\max Q_2(C)$ as described in Section 4.3 ; /* Update IBP Weights */
  /* Rotation-Step */
  $A = \frac{1}{N\cdot M}\sum_{i=1}^{N}\sum_{m=1}^{M}\theta_i^{(m)}\theta_i^{(m)'}$; $B = B * A_L$, where $A_l A_L' = A$;
end

- For any n-dimensional vector v, let $\{v = 0\}$ denote the n-dimensional binary vector, in which the ith component is 1 iff $v_i = 0$. Let

$$D_1 = \text{diag}(2Y_{J_B} - \mathbb{1}_{J_B})B_{J_B}, \quad D_2 = \text{diag}(2Y_{J_B} - \mathbb{1}_{J_B})D_{J_B},$$
$$D_3 = \text{diag}(2\{Y_{J_{O_2}} = 0\} - \mathbb{1}_{J_{O_2}})B_{J_{O_2}},$$
$$D_4 = \text{diag}(2\{Y_{J_{O_2}} = 0\} - \mathbb{1}_{J_{O_2}})D_{J_{O_2}},$$

  where $D_{J_{O_2}}$ has its jth component as $d_{j,0}$ when $y_j = 0$, and as d_{j,L_j-1} when $y_j = L_j$.
- Let $B_{J_{O_1}}^{(i)}$ be the ith row vector of $B_{J_{O_1}}$, and $D_j^{(i)}$ be the ith component of the intercept vector of the jth feature in J_{O_1}. Define

$$\tilde{B}_{J_{O_1}} = [B_{J_{O_1}}^{(1)'}, -B_{J_{O_1}}^{(1)'}, \ldots, B_{J_{O_1}}^{(|J_{O_1}|)'}, -B_{J_{O_1}}^{(|J_{O_1}|)'}]' \in \mathbb{R}^{2|J_{O_1}|\times K},$$
$$\tilde{D}_{J_{O_1}} = [D_1^{(y_j)}, -D_1^{(y_j-1)} \cdots, D_{|J_{O_1}|}^{(y_j)}, -D_{|J_{O_1}|}^{(y_j-1)}] \in \mathbb{R}^{2|J_{O_1}|}.$$

- Let $\overline{\mathbb{I}_{\mathbb{J}_{\mathbb{O}_1}}}$ be a $2|J_{O_1}|$ by $2|J_{O_1}|$ block diagonal matrix, each block of which is the 2 by 2 matrix $-\mathbb{1}_2\mathbb{1}_2' + 2\mathbb{I}_2$.

We are now ready to characterize the conditional posterior distribution for each latent factor θ_i as an instance of the unified skew-normal distribution:

Theorem 5.1. Suppose the model parameters (B, D, Σ) are known. Let $y = (y_1, \ldots, y_J)' \in \mathbb{R}^J$ be a vector of conditionally independent response with mixed data types, each component of which is generated through either equations (1), (5), or (6). Assuming latent factor prior $\theta_i \sim N(\xi, \Omega) \in \mathbb{R}^K$. Then

$$(\theta_i \mid y, B, D, \Sigma) \sim \text{SUN}_{K,\bar{J}}\left(\xi_{\text{post}}, \Omega_{\text{post}}, \Delta_{\text{post}}, \gamma_{\text{post}}, \Gamma_{\text{post}}\right)$$

with posterior parameters

$$\xi_{\text{post}} = \Omega_{\text{post}}[\tilde{B}_{J_c}'\tilde{B}_{J_c}\hat{\theta}_i + \Omega^{-1}\xi],$$
$$\Omega_{\text{post}} = (\tilde{B}_{J_c}'\tilde{B}_{J_c} + \Omega^{-1})^{-1},$$
$$\Delta_{\text{post}} = \bar{\Omega}_{\text{post}}\,\omega_{\text{post}}\,\bar{D}'S^{-1},$$
$$\gamma_{\text{post}} = S^{-1}(\bar{D}\xi_{\text{post}} + \bar{V}),$$
$$\Gamma_{\text{post}} = S^{-1}\left(\bar{D}\Omega_{\text{post}}\bar{D}' + \overline{\mathbb{I}_{\mathbb{J}}}\right)S^{-1},$$

where $\tilde{B}_{j_c} = \Sigma^{-\frac{1}{2}}B_{J_c}$, $\hat{\theta}_i = (\tilde{B}_{J_c}'\tilde{B}_{j_c})^{-1}\tilde{B}_{J_c}'\Sigma^{-\frac{1}{2}}y_{J_c}$, $\bar{J} = |J_B| + 2|J_{O_1}| + |J_{O_2}|$, $\bar{D} = [D_1', D_3', \tilde{B}_{J_{O_1}}']' \in \mathbb{R}^{\bar{J}\times K}$ with $\overline{D_j} \in \mathbb{R}^k$ as its jth row,

$$S = \text{diag}\{(\overline{D_1}'\Omega_{\text{post}}\overline{D_1} + 1)^{\frac{1}{2}}, \ldots, (\overline{D_{\bar{J}}}'\Omega_{\text{post}}\overline{D_{\bar{J}}} + 1)^{\frac{1}{2}}\} \in \mathbb{R}^{\bar{J}\times\bar{J}},$$

$\bar{V} = [D_2', D_4', \tilde{D}_{J_{O_1}}']'$, and $\overline{\mathbb{I}_{\mathbb{J}}}$ is a $\bar{J}$ by $\bar{J}$ diagonal block matrix, in which $\overline{\mathbb{I}_{\mathbb{J}}}_{[1,1]} = \mathbb{I}_{|\mathbb{J}_{\mathbb{B}}|+|\mathbb{J}_{\mathbb{O}_2}|}$ and $\overline{\mathbb{I}_{\mathbb{J}}}_{[2,2]} = \overline{\mathbb{I}_{\mathbb{J}_{\mathbb{O}_1}}}$.

Theorem 5.1 is particularly valuable because it demonstrates that sampling from the latent factor posterior distribution remains tractable under mixed item response data types, as it still conforms to an instance of the unified skew-normal distribution. This ensures that our methodology can be generalized to handle a mix of continuous, binary, and ordinal item response types. The proof of Theorem 5.1, and the comprehensive EM procedure are provided in Appendix I.

6. Experiments

6.1. Dynamic Posterior Exploration of EM

A common critique of the Bayesian approach centers on the perceived arbitrariness in prior specification and the challenges in prior calibration. Here we introduce a systematic methodology to calibrate the priors in our EM algorithm, aiming to achieve optimal outcomes. The calibration of penalty parameters λ_0 and λ_1 of the SSL prior is of paramount importance for effectively exploring the posterior landscape. In Ročková and George (2016), the authors adopt the idea of deterministic annealing Ueda and Nakano (1998) and Yoshida and West (2010) and propose to keep the value of λ_1 fixed at a reasonable level, and initialize λ_0 at a small value to quickly obtain an initial solution. They then iteratively refit the model by gradually increasing the value

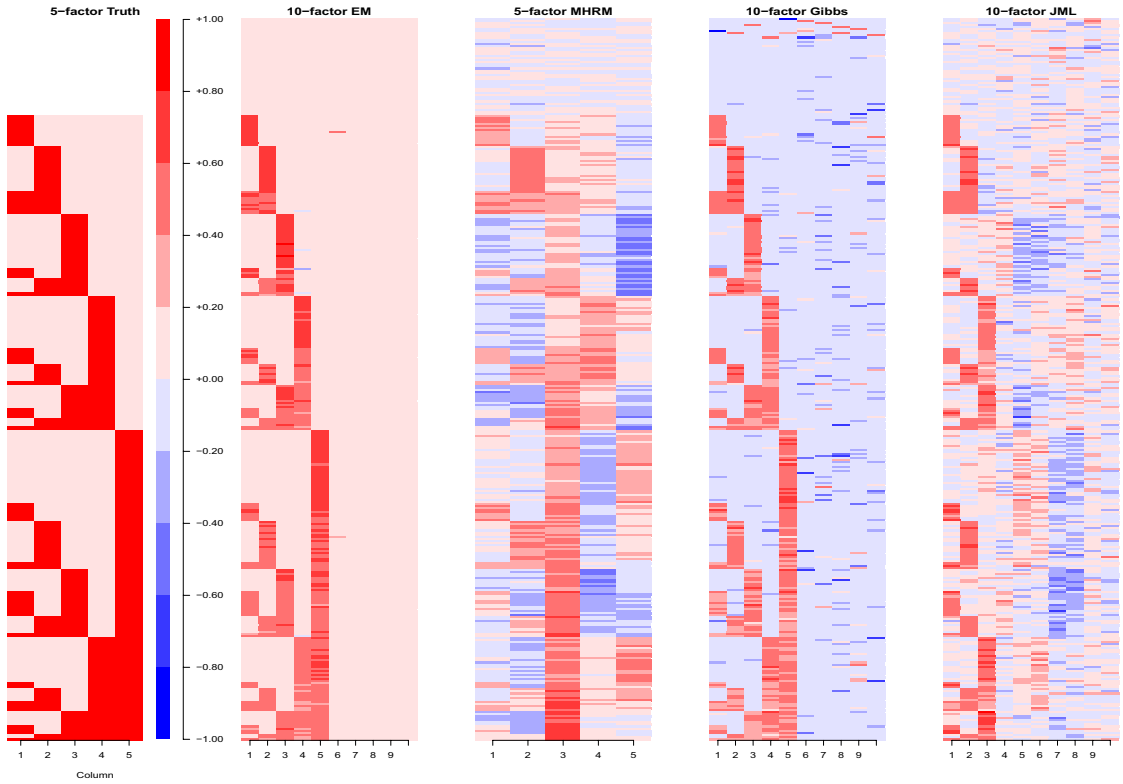


Figure 2. IBP loading matrix estimation: baseline models.

of λ_0 and reinitializing the model parameters from the previous iteration. Once the increase of λ_0 no longer affects the solution, the model has stabilized and its solution is ready for interpretation. We can readily apply this approach of "dynamic posterior exploration" to our PXL-EM algorithm for MIRT models. In Section 6.2, we will describe this calibration process of in detail.

6.2. IBP Loading Matrix

We randomly generated the true loading matrix from the Indian Buffet Process (IBP) prior with an intensity parameter $\alpha = 2$, resulting in a complex loading structure as shown in the leftmost subplot of Figure 2. Since a draw from IBP prior would be infinite-dimensional, we applied a stick-breaking approximation to keep the strongest five factors, and reordered the rows and columns of the loading matrix to enhance visualization. We focused on a difficult high-dimensional scenario, with the number of observations $N = 250$ being smaller than the number of items $J = 350$.

To assess whether PXL-EM can recover the true number of factors, we deliberately overshot the number of factors as $K^* = 10$. Keeping $\lambda_1 = 0.5$ fixed, we gradually increased λ_0 along the path $\{0.5, 1, 3, 6, 10, 20, 30, 40\}$ following the procedure described in Section 6.1. A visualization of the loading matrix estimation along this λ_0-path can be found in Appendix E.1. Convergence was determined when $|B^{(t+1)} - B^{(t)}|_\infty < 0.06$. Table 1 documents how the estimation accuracy dramatically improved over time.

We also considered the MHRM algorithm, the sparse Gibbs sampler, and the JML estimation method as introduced in Section 2. In Figure 2, we observe that the Gibbs sampler was able to recover the general structure of the factor loading, but required further thresholding to output exact zeros. On the other hand, the MHRM algorithm suffered from rotational indeterminacy issues due to the absence of structural assumptions on the sparsity of the loading matrix. The JML estimation method converged amazingly fast compared to the other methods. While it captured the general IBP pattern, it only identified four factors, and missed several major components for each of the recovered factor.

As shown in Table 2, our PXL-EM algorithm achieved the best overall performance in the accuracy of loading estimation, as well as the recovery of true zeros. One might have noticed that PXL-EM was not dramatically fast in this experiment. It is essential to realize our PXL-EM had already attained matching

Table 1. Dynamic posterior exploration of PXL-EM.

λ_0	Loading MSE	Intercept MSE	FDR	FNR	Accumulated time (s)
1	0.078	0.035	0.666	0	554
6	0.016	0.028	0.544	0	1433
20	0.011	0.022	0.103	0	2173
30	0.01	0.022	0.022	0	2450
40	0.01	0.022	0.009	0	2682

Table 2. Comparison of model performances.

Algorithm	Loading MSE	Intercept MSE	FDR	FNR	Time (sec)
PXL-EM	0.01	0.022	0.009	0.000	2682
10-factor Gibbs	0.018	0.037	0.542*	0.000*	2385
JML	0.165	0.079	NA	NA	4
MHRM	0.214	0.043	NA	NA	1848

*FDR and FNR for Gibbs samplers were computed after thresholding.

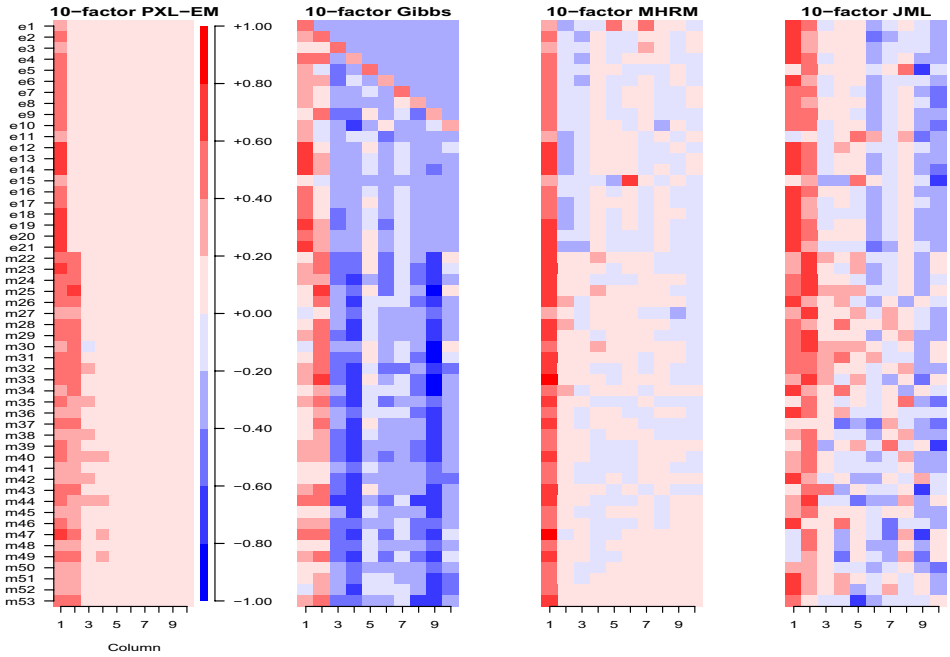


Figure 3. DESE loading matrix estimation.

performance when $\lambda_0 = 6$. The Gibbs sampler would not scale well with either the sample size or the truncated dimension. The availability of more CPUs can always effectively accelerate our EM algorithm, but may not speed up the Gibbs sampler. As the sample size N increases, we can expect the estimation becomes increasingly accurate. This is due to the IBP loading structure possessing a GLT structure, making it identifiable by Theorem 3.3. In Appendix E.2, we empirically verify consistency by repeating the same experiment with varying N from 100 to 1000.

6.3. Massachusetts DESE Data

To highlight the advantages of our approach in educational assessment, we used student item response data from Massachusetts Department of Elementary and Secondary Education (DESE). The DESE datasets encompass students' item response to both the English and the Mathematics exam items administrated in Massachusetts in 2022. We focus on the grade 10 students who participated in both the English and Math exams, resulting in a total sample size $N = 60{,}918$. We omitted items with unconventional scores, such as the writing scores in the English exams for simplicity. This resulted in a dataset with 21 English items and 32 math items eligible for analysis.[1]

Considering that a certain level of English skill is necessary to solve math items, but not necessarily vice versa, we anticipated that math items load on both the English and math factors. However, the math factors for the English items should be negligible. To assess whether our approach would successfully recover such a sparse structure, we fit the DESE dataset with our PXL-EM algorithm, and set the truncated dimension as 10. Setting $\lambda_1 = 0.1$, we deployed the procedure of dynamic posterior exploration with convergence claimed when $|B^{(t+1)} - B^{(t)}|_\infty < 0.04$. Despite the large sample size, the PXL-EM algorithm converged in just 3483 sec, whereas the competing Bayesian method using Gibbs sampling took over 15 hr. This highlights the computational efficiency of our E-step in large N setting. For the frequentist approaches, the JML and MHRM estimation algorithms took 2055 and 8225 sec, respectively, to fit the data.

Figure 3 presents the estimated factor loading matrix for each model. We've labeled each row to indicate its corresponding item type as either English or math. Notably, our PXL-EM algorithm demonstrates a distinct pattern: high factor loadings for the secondary factor commence at item m22, aligning precisely with our expectations. Impressively, all math items exhibit substantial positive loadings on the second factor, yet the second factor for the English items are negligible, rendering our estimated loading matrix considerably more interpretable compared to the other models. In contrast, both the Gibbs sampler and the JML estimation algorithm managed to identify two prominent components, but neither of them successfully captured the distinction between the math and the english items. More importantly, for all the competing exploratory methods, their loading estimates for the extra dimensions 3–10 significantly deviate from zero, making it very difficult to determine either the number of factors, or the potential latent structure.

Observant readers might notice that our estimated loadings for the math items are noisier than the English ones, and are not entirely zero across dimensions 3–4. This is intuitive as DESE classifies these 32 math items into four categories: algebra (13 items), geometry (13 items), number & quantity (3 items) and statistics (3 items). There were only 11 nonzero estimates out of 32 total math items for the fourth factor, among which only 4 items exhibit loadings higher than 0.2. Interestingly, all three rare statistics items have loadings higher than 0.2 on the fourth factor, which cannot be coincidental. These intriguing patterns suggest that the extra factors might be associated with

[1]Exam questions are available here: *https://www.doe.mass.edu/mcas/2022/release/*.

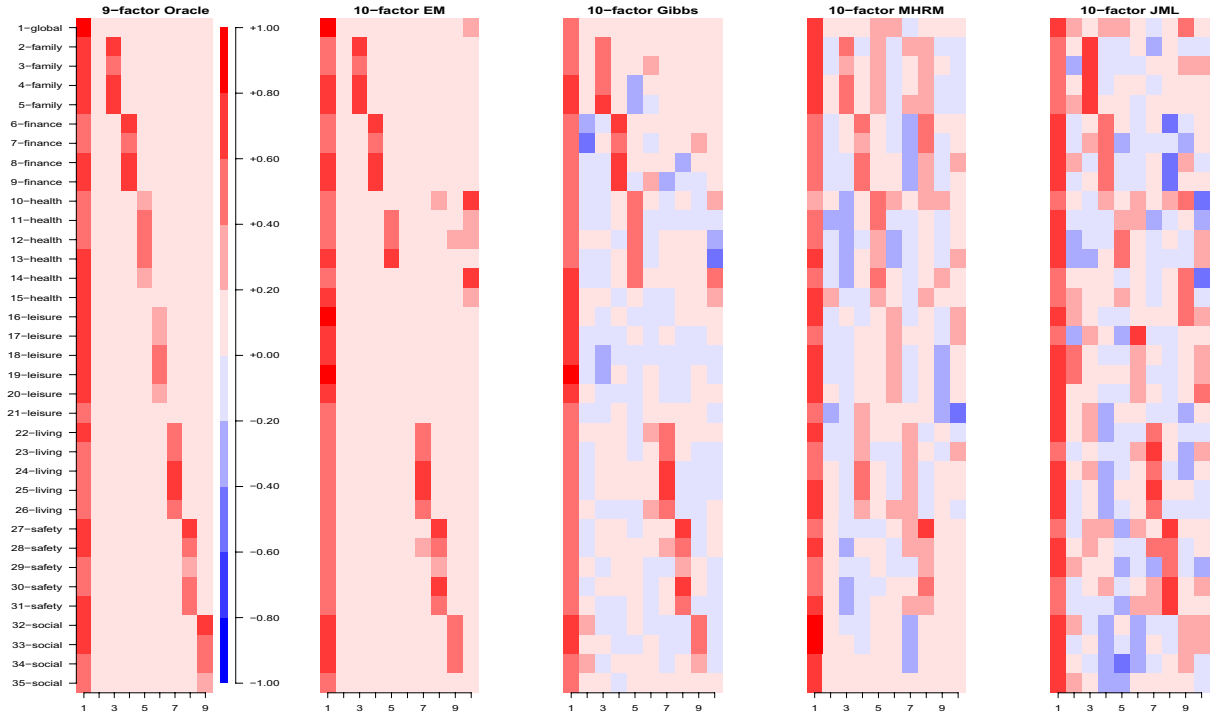


Figure 4. QOL loading matrix estimation.

the detection of rare skills, rather than simply being an artifact of estimation noise. Tables of exact factor loading estimates for each item can be found in the Appendix F.

This experiment emphasizes the benefits of our proposed exploratory MIRT framework in producing interpretable and sparse solutions, as well as detecting the nuances of rare latent math skills in a large N setting. Our method eliminates the need for expert item categorization into rigid subdomains, and is able to make interesting and objective discoveries from data that might be unaware by experts.

6.4. Quality of Life Measurement

We demonstrate the potential of our approach by applying it to a more intricate Quality of Life (QOL) dataset discussed in Lehman (1988). This dataset comprises responses from 586 individuals with chronic mental illnesses, who answered 35 items related to their life satisfaction across seven distinct subdomains (family, finance, health, leisure, living, safety, and social). In the psychological measurement literature, Gibbons et al. (2007) showcased the advantages of the bifactor model in a confirmatory analysis. We replicated their analysis, and the estimated bifactor loading structure is visualized in the leftmost plot of Figure 4.

Our goal is to determine whether our unconstrained PXL-EM algorithm can reveal a similar bifactor structure without any confirmatory guidance. This particular instance presents a significant challenge, as the algorithm must learn an eight-dimensional latent space from a mere 35 available items. We once again adopted an overshot latent dimension of $K^* = 10$ and conducted dynamic posterior exploration. In Figure 4, we observe a remarkable resemblance between our estimated latent structure and the bifactor loading pattern. Our algorithm successfully identifies the primary factor and accurately recovers 6 out of the 7 subdomains, only with the exception of the "Leisure" subdomain. It also correctly discerns the actual latent dimensionality, as dimensions 2 and 6 are entirely zeros.

The loading matrix produced by our PXL-EM algorithm only differs from the oracle bifactor structure in two subdomains: the additional dimensionality for the "Health" items, and the absence of "Leisure" factor account for the FDR and FNR, respectively as shown in Table 3. In contrast, both the MHRM algorithm and the Gibbs sampler used all 10 dimensions, hence, overfitting the item response data and having a marginally lower in-sample reconstruction MSE. The JML estimation method converged dramatically fast for this small N dataset, but obtained relatively high estimation MSE, and a noisy loading representation. In contrast, the pink areas visualized by our PXL-EM algorithm are exact zeros. We provide the detailed loading matrices and more discussions in Appendix G.

Essentially, if the goal of exploratory analysis is to cluster items into distinct types, our PXL-EM algorithm clearly demonstrates distinct characteristics among all seven subdomains. This experiment showcases the power of our proposed EM approach to learn high-dimensional latent structures with very few items, highlighting its unique advantages compared to other state-of-the-art exploratory factor analysis strategies.

6.5. Ordinal Bio-Behavioral Data

To highlight the potential of our methodology in uncovering new scientific insights, we apply it to a challenging ordinal bio-behavioral dataset with intricate latent structures. Recent research Stan et al. (2020) used the confirmatory MIRT bifactor model to identify strong negative correlations between corti-

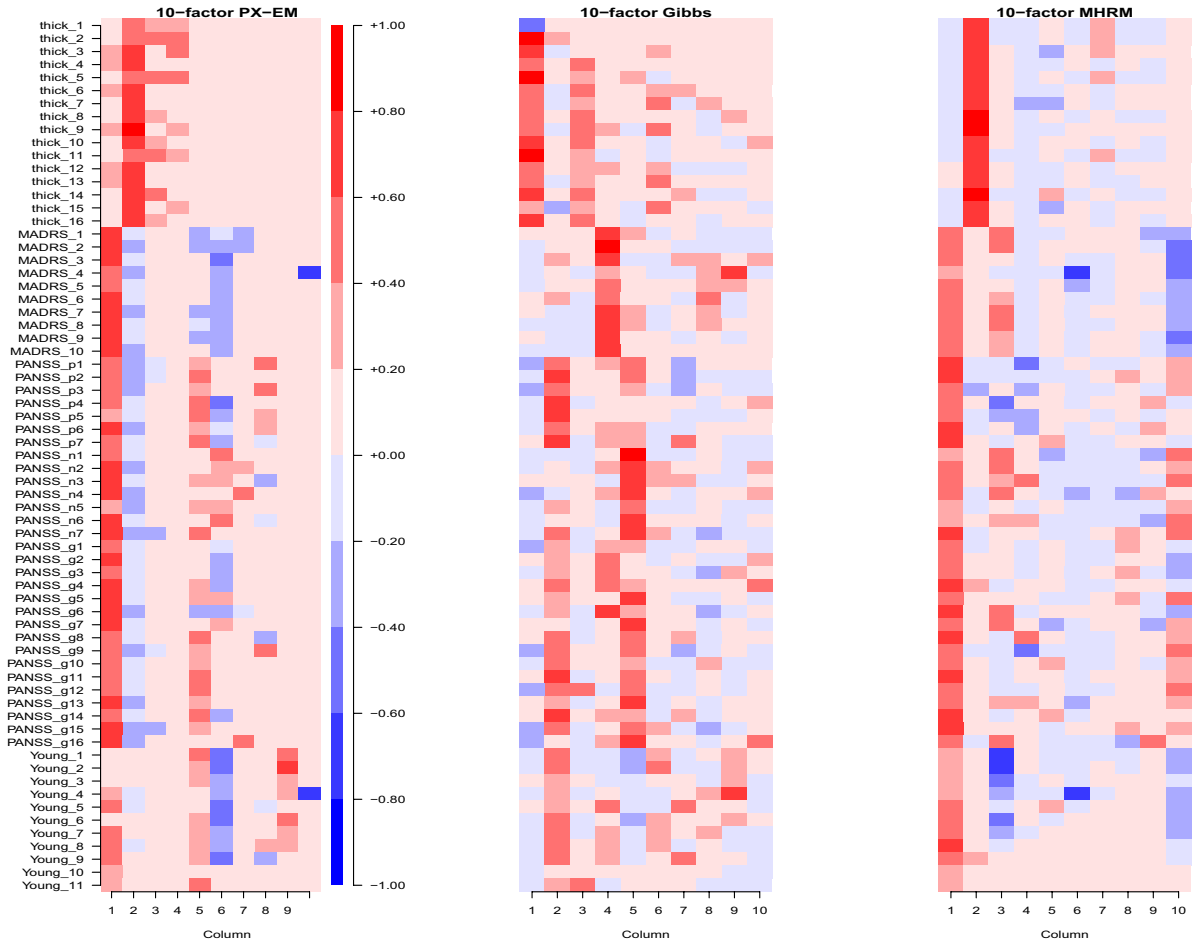


Figure 5. Estimated bio-behavioral factor loading matrices.

Table 3. Model performances for QOL data.

Algorithm	FDR*	FNR*	Reconstruction MSE	Time (sec)
PXL-EM	0.092	0.157	0.102	411
Sparse Gibbs	0.678	0.057	0.096	410
MHRM	NA	NA	0.088	350
JML	NA	NA	0.156	3

*We performed thresholding for the Gibbs sampler, but not for the PXL-EM.

cal thickness and severity of clinical symptoms. Our goal is to determine whether our exploratory data-driven framework can make similar discoveries without the need to prespecify a loading structure and the number of factors, which typically require multiple stages of statistical analysis in a confirmatory setting.

Following the analysis in Stan et al. (2020), we investigate the relationship between the 16 cerebral cortex regions (ordinal scale 1–5) and 51 symptom items across 5 subdomains: MADRS (depression; 10 items), PANSS-P (positive psychosis; 7 items), PANSS-N (negative psychosis; 7 items), PANSS-G (general psychosis; 16 items), and Young (mania; 11 items). The symptom items vary in ordinal scales from 1 to 8. We consider participants who have both complete cortical thickness items and clinical symptom items, resulting in a sample size of $N = 575$. We set truncation level $K^* = 10$, and deploy our PXL-EM algorithm via dynamic posterior exploration. The estimated loading matrix along with the ones produced by the other competing methods are presented in Figure 5.[2]

As indicated by the color palette, the red, light pink, and blue regions represent negative, zero, and positive loadings, respectively. Notably, the loading matrix estimated by PXL-EM is significantly sparser and more interpretable compared to those produced by the other methods: the first 1–3 dimensions represent the primary bio-behavioral dimensions of interest, shared by both the cortical thickness items (first 16 rows) and the clinical symptom items (last 51 rows). Dimension 4 is an independent factor for cortical thickness, while dimensions 5–10 are shared by the 51 clinical symptoms.

To investigate the relationship between cortical thickness and clinical psychotic manifestations, it is essential to identify the (bio-behavioral) factors that are shared by all the items. Such analysis is unfeasible based on the factor loadings produced by the Gibbs sampler and the MHRM algorithm, where all the items are associated with all the 10 factors, because these methods neither produce exact zeros nor can determine the number of factors. In contrast, our PXL-EM algorithm simplifies this investigation by identifying only three bio-behavioral dimensions, with the remaining seven dimensions independently loaded by either the biological or clinical features. More importantly, the

[2]We did not include the JML method this time because it does not seem to handle ordinal datasets.

second bio-behavioral dimension clearly highlights the negative correlation between cortical thickness and the severity of clinical symptoms, as the cortical thickness items are predominantly red, while the clinical symptoms items are overwhelmingly blue.

In addition to identifying the negative correlation, we are also able to pinpoint the exact symptoms that are highly correlated with reduced cortical thickness. The research in Stan et al. (2020) highlights seven symptom items with highly negative factor loadings lower than −0.25: delusions (PANSS-p1), hallucinatory behavior (PANSS-p3), suspiciousness (PANSS-p6), depression (PANSS-g6), unusual thought content (PANSS-g9), active social avoidance (PANSS-g16), and passive-apathetic social withdrawal (PANSS-n4). Remarkably, by examining the second bio-behavioral dimension, we identify six symptoms with factor loadings lower than −0.25, each of which is among the seven highlighted symptoms listed above. The only missing item is PANSS-g6, which still has a very negative loading of −0.24. We invite readers to refer to the tables in Appendix H for the detailed estimates and comparison.

This experiment demonstrates the generality and effectiveness of our approach in handling complex bio-behavioral ordinal datasets with nuanced latent structures. Not only can we generate a sparse and interpretable solution unattainable by other competing estimation methods, but our scientific discoveries also closely align with the established research findings. Moreover, our exploratory data-driven approach avoids making assumptions about the loading structure, which is particularly valuable in settings where such assumptions are too ambiguous to be made, as is often the case in scientific inquiries.

7. Discussion

This article advocates for a novel Bayesian framework for estimating exploratory sparse MIRT models. The combination of the Indian Buffet Process prior and the spike-and-slab LASSO prior allows our EM algorithm to simultaneously learn latent dimensionality and sparse representations of factor loadings, which have not been sufficiently addressed in the categorical actor analysis literature. The efficiency of the algorithm is driven by the adaptation of parameter expansion, as well as by an efficient sampling strategy during the E-step, where we explicitly derived latent factor posterior distributions. We further demonstrate the advantage of adopting a probit link for establishing identification results and provide theoretical guarantees for recovering the true number of factors and contraction toward the true factor loading matrix under our prior specification. Our approach is capable of accommodating mixed data types, and we showcase the effectiveness of our approach on a challenging synthetic dataset and three real-world datasets from different domains. In each experiment presented in Section 6, our algorithm consistently generates sparse and interpretable estimations of the factor loading matrices, demonstrating its unique strength compared to other popular exploratory MIRT estimation frameworks.

While we focus on the estimation of the factor loadings and intercepts, our framework also has direct implications for latent trait measurements. For a well-calibrated set of J items where the item parameters are known, Theorems 4.2 essentially provides an analytic form of the latent factor posterior distribution of any new observation $y \in \mathbb{R}^J$. This avoids the additional computational need to perform MCMC sampling in order to conduct posterior inference on latent traits. Compared to the other MIRT estimation strategies, this can potentially save tremendous amount of computational time when the number of observations is large. Furthermore, many important properties of the unified skew-normal distribution such as the mean and covariance are readily available in Arellano-Valle and Azzalini (2022), making exact Bayesian inference possible. For more complicated objects of inferential interest, one may also consider the efficient posterior sampling strategies outlined in Section 4.2 to compute credible intervals. In Appendix G.2, we conducted a comparative analysis of the estimated latent factors and their posterior variances between our proposed PXL-EM approach and the oracle bifactor model for the QOL dataset, highlighting the potential of our approach in posterior factor inference and psychological measurements.

Several promising research directions can shape the future of our approach. First, advancements in evaluating high-dimensional Gaussian cumulative distribution functions could eliminate the need for Monte-Carlo simulations in the E-step and factor rotation step. This improvement would not only speed up computation but also simplify Bayesian inference of latent factors following factor loading estimation. Moreover, we can enhance the flexibility of our Bayesian model. Recent work by Anceschi et al. (2023) has demonstrated posterior conjugacy in probit and tobit models, allowing the prior for coefficient parameters to follow a unified-skew normal distribution (SUN). This adjustment could further refine the modeling capabilities of our approach, with implications for various applications.

While our methodology can handle mixed continuous, binary, and ordinal responses, it would be interesting to extend it to other item types such as count data, or to embed our Bayesian model into generalized mixed-models for longitudinal data. One can derive the latent factor posterior distribution using techniques similar to those in Theorem 4.2, in order to create a feasible E-step to estimate various factor models of interest.

Supplementary Materials

The online supplementary materials include all technical proofs, along with figures and tables corresponding to the experiments presented in Section 6. An official R implementation of the proposed EM algorithm is available at *https://github.com/JiguangLi/sparse_bayesian_mirt*.

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Disclosure Statement

The authors report there are no competing interests to declare.

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